CDA 106
Numerical Methods for Data Science

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Elementary matrices

$$\underbrace{E_n \cdots E_2 E_1}\, A = U$$

$$\underline{A = LU}$$

Consider the matrix

$$A = \begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & -2 \\ -1 & 2 & 1 \end{bmatrix}$$

The above matrix A has all the principle minors as non-zero.

$$|2| \neq 0 \qquad \begin{vmatrix} 2 & 5 \\ 3 & 1 \end{vmatrix} \neq 0 \qquad \begin{vmatrix} 2 & 5 & 3 \\ 3 & 1 & -2 \\ -1 & 2 & 1 \end{vmatrix} \neq 0$$

<u>Aim</u>: To find LU decomposition via elementary matrices (elementary row operations)

$$A = \begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & -2 \\ -1 & 2 & 1 \end{bmatrix}$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$E_1 A = B$

$$\underbrace{R_2 - \tfrac{3}{2}R_1}\ \begin{bmatrix} 2 & 5 & 3 \\ 0 & -\frac{13}{2} & -\frac{13}{2} \\ -1 & 2 & 1 \end{bmatrix} \quad (= B)$$

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix}$$

$E_2 B = C$

$$\underbrace{R_3 + \tfrac{1}{2}R_1}\ \begin{bmatrix} 2 & 5 & 3 \\ 0 & -\frac{13}{2} & -\frac{13}{2} \\ 0 & 9/2 & 5/2 \end{bmatrix} \quad (= C)$$

$$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 9/13 & 1 \end{bmatrix}$$

$E_3 C = D$

$$\underbrace{R_3 + \tfrac{9}{13}R_2}\ \begin{bmatrix} 2 & 5 & 3 \\ 0 & -\frac{13}{2} & -\frac{13}{2} \\ 0 & 0 & -2 \end{bmatrix} \quad (= D) = U$$

Note $\underline{E_3 E_2 E_1 A = U}$

$$E_3 E_2 E_1 A = U$$

$$\Rightarrow \quad A = \underbrace{E_1^{-1} E_2^{-1} E_3^{-1}}_{L} U$$

<u>Imp. Note</u>: ① $L_1 L_2 = L_3$

② $L_1 L_2 \cdots L_n = L$

③ $U_1 U_2 \cdots U_n = U$

Result Suppose E_k is elementary matrix obtained by (of I_3) adding k times a row to another row, i.e. (for example)

say, $R_2 \to R_2 + kR_1$

$$E_k = \begin{bmatrix} 1 & 0 & 0 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{then} \quad E_k^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -k & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Proof: $|E_k| = 1$

$$E_k = \begin{array}{c} \begin{array}{ccc} a_{11} & a_{12} & a_{13} \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} = A$$

$a_{11} = a_{22} = a_{33} = 1$, $a_{21} = k$

$a_{ij} = 0$ remaing

$$\text{Coff. of } E_k = \begin{bmatrix} 1 & -k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Coff. of $a_{11} = 1 = (-1)^{1+1} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$

" $a_{22} = 1$

" $a_{33} = 1$

Coff. of $a_{12} = (-1)^{1+2} \begin{vmatrix} k & 0 \\ 0 & 1 \end{vmatrix} = -k$

$$E_k^{-1} = \frac{\text{Adj } E_k}{|E_k|} = \frac{(\text{Coff } E_k)^T}{1} = \begin{bmatrix} 1 & 0 & 0 \\ -k & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -3/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1 \end{bmatrix} \qquad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 9/13 & 1 \end{bmatrix}$$

$$E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad E_2^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1/2 & 0 & 1 \end{bmatrix} \qquad E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -9/13 & 1 \end{bmatrix}$$

Therefore $E_1^{-1} E_2^{-1} E_3^{-1} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 3/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1/2 & 0 & 1 \end{bmatrix}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -9/13 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 3/2 & 1 & 0 \\ -1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -9/13 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3/2 & 1 & 0 \\ -1/2 & -9/13 & 1 \end{bmatrix} = L$$

Algorithm for LU decomposition of a matrix A

① Use row operations to reduce matrix A to REF (U)

[Do not use row exchange operation]

(upper △ matrix)

② Keep track of all the $\boxed{\text{multipliers}}$ used to create zeros below the pivot element.

③ Now, at each position in an Identity matrix below the diagonal elements ("1"), place the negative of multipliers used in step ② in that position.

④ You will get L matrix in step ③ & ~~thereof~~ thereby

$$A = LU.$$

Example

$$A = \begin{bmatrix} 2 & 4 & -6 \\ 1 & 5 & 3 \\ 1 & 3 & 2 \end{bmatrix}$$

$$\underbrace{R_2 - \tfrac{1}{2}R_1}_{R_3 - \frac{1}{2}R_1} \begin{bmatrix} 2 & 4 & -6 \\ 0 & 3 & 6 \\ 0 & 1 & 5 \end{bmatrix}$$

$$\underbrace{R_3 - \tfrac{1}{3}R_2}_{} \begin{bmatrix} 2 & 4 & -6 \\ 0 & 3 & 6 \\ 0 & 0 & 3 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \boxed{1/2} & 1 & 0 \\ 1/2 & 1/3 & 1 \end{bmatrix}$$

$$\boxed{A = LU}$$

Now, consider

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 2 & 5 & 8 \end{bmatrix}$$

$$\xrightarrow[R_3 - \frac{1}{2}R_1]{R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 3 & 6 \end{bmatrix}$$

This is your pivot which is zero

In this case we cannot proceed further as per the algorithm discussed for LU decomposition.

The LU decomposition is not possible, however we can find PLU decomposition.

PLU decomposition

For a given matrix A, we need to find 3 matrices, P, L, U s.t.

$$PA = LU$$

This means that in this case, to get LU decomposition, we need to do some row exchanges in matrix A.

Interesting property ① Each permutation matrix is invertible.

② $P^{-1} = P^T$

Consider the PLU decomposition of

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 2 & 5 & 8 \end{bmatrix}$$

E_1

$R_2 - R_1$

$R_3 - 2R_1$

E_2

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 3 & 6 \end{bmatrix} \quad E_2 E_1 A$$

$R_2 \longleftrightarrow R_3$

P

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 6 \\ 0 & 0 & 2 \end{bmatrix} \quad P E_2 E_1 A$$

U

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$P = \begin{array}{c} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \end{array} \begin{array}{c} R_1 \\ R_3 \\ R_2 \end{array}$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Please do take due care here!!

In this way we found $\boxed{PA = LU}$ ✓

$$LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 8 \\ 1 & 1 & 3 \end{bmatrix}$$

$$= PA$$